Quantum Gravitational Dynamics

Chapter 6: Exact Solutions and the Post-Newtonian Hierarchy

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Abstract

We construct exact algebraic metrics for arbitrary configurations of Kerr bodies. The σ-field superposition principle, proved with rigorous error bounds, yields closed-form N-body spacetimes whose cross-terms automatically encode binding energy, gravitational waves, and frame-dragging. The post-Newtonian expansion is derived to 6PN using the master formula $e_n^{(\eta=0)} = -\binom{2n}{n}(3/4)^n(2n-1)/(n+1)$, which generates all test-body binding energy coefficients as exact rational numbers. Three closed-form spinning series extend these results to arbitrary spin-orbit and spin-spin orders. We catalogue all QGD-unique contributions to the gravitational wave flux through 6PN, including a dipole term at -1PN and a dipole tail at 0.5PN that has no GR analogue.

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1 Single-Body σ-Fields

For a Kerr body of mass M_a, spin parameter $\alpha_a = J_a/(M_a c)$, located at $\mathbf{x}_a(t)$, the two nonzero σ-components at field point $\mathbf{x}$ are:

$$\sigma_t^{(a)}(\mathbf{x}) = \sqrt{\frac{2GM_a\, r_a}{c^2\, \mathcal{S}_a}}, \qquad r_a = |\mathbf{x} - \mathbf{x}_a|, \qquad \mathcal{S}_a = r_a^2 + \alpha_a^2 \cos^2\theta_a \tag{1}$$

$$\sigma_\phi^{(a)}(\mathbf{x}) = \alpha_a \sin\theta_a \sqrt{\frac{2GM_a}{c^2\, r_a\, \mathcal{S}_a}} \tag{2}$$

The product relation gives the frame-dragging amplitude directly:

$$\sigma_t^{(a)} \cdot \sigma_\phi^{(a)} = \frac{2GM_a \alpha_a \sin\theta_a}{c^2\, \mathcal{S}_a} \tag{3}$$

For a non-spinning Schwarzschild body ($\alpha_a = 0$):

$$\sigma_t^{(a)} = \sqrt{\frac{2GM_a}{c^2\, r_a}}, \qquad \sigma_\phi^{(a)} = 0 \tag{4}$$

The master metric gives, for each single body, exactly the Boyer–Lindquist Kerr metric. All standard solutions (Schwarzschild, Reissner–Nordström, Kerr–Newman, Schwarzschild–de Sitter) follow from the source signature table: mass ($\varepsilon = +1$), charge ($\varepsilon = -1$), spin ($\varepsilon = +1$), $\Lambda > 0$ ($\varepsilon = +1$), $\Lambda < 0$ ($\varepsilon = -1$).

2 The Two-Body Problem

2.1 Field superposition

With bodies $a = 1, 2$, the total fields are:

$$A_{\text{tot}} \equiv \sigma_t^{(\text{tot})} = \sigma_t^{(1)} + \sigma_t^{(2)}, \qquad B_{\text{tot}} \equiv \sigma_\phi^{(\text{tot})} = \sigma_\phi^{(1)} + \sigma_\phi^{(2)} \tag{5}$$

2.2 Complete two-body metric

Applying the master equation with $T = \mathrm{diag}(1, 1, r, r\sin\theta)$, the exact metric in the equatorial plane is:

$$g_{tt} = -\left(1 - (\sigma_t^{(1)} + \sigma_t^{(2)})^2\right) \tag{6}$$

$$g_{t\phi} = -r\sin\theta\,(\sigma_t^{(1)} + \sigma_t^{(2)})(\sigma_\phi^{(1)} + \sigma_\phi^{(2)}) \tag{7}$$

$$g_{\phi\phi} = r^2\sin^2\theta\left(1 + (\sigma_\phi^{(1)} + \sigma_\phi^{(2)})^2\right) \tag{8}$$

$$g_{rr} = \frac{1}{1 - (\sigma_t^{(1)} + \sigma_t^{(2)})^2} \tag{9}$$

2.3 Cross-term expansion: g_{tt}

Expanding the square:

$$\boxed{g_{tt} = -\left[1 - \underbrace{(\sigma_t^{(1)})^2}_{2GM_1/(c^2 r_1)} - \underbrace{(\sigma_t^{(2)})^2}_{2GM_2/(c^2 r_2)} - \underbrace{2\sigma_t^{(1)}\sigma_t^{(2)}}_{\text{QGD cross}}\right]} \tag{10}$$

The cross-term evaluates to:

$$2\sigma_t^{(1)}\sigma_t^{(2)} = \frac{4G\sqrt{M_1 M_2}\sqrt{r_1 r_2}}{c^2\sqrt{\mathcal{S}_1\mathcal{S}_2}} \xrightarrow{\alpha_i = 0} \frac{4G\sqrt{M_1 M_2}}{c^2\sqrt{r_1 r_2}} \tag{11}$$

This scales as $r^{-1/2} \cdot r^{-1/2} = r^{-1}$ — a distinct gravitational coupling between the two mass fields.

2.4 Cross-term expansion: $g_{t\phi}$ (four spin-coupling terms)

$$g_{t\phi} = -r\sin\theta\Big[\underbrace{\sigma_t^{(1)}\sigma_\phi^{(1)}}_{\text{body 1 frame-drag}} + \underbrace{\sigma_t^{(2)}\sigma_\phi^{(2)}}_{\text{body 2 frame-drag}} + \underbrace{\sigma_t^{(1)}\sigma_\phi^{(2)}}_{\text{cross: } M_1\times\alpha_2} + \underbrace{\sigma_t^{(2)}\sigma_\phi^{(1)}}_{\text{cross: } M_2\times\alpha_1}\Big] \tag{12}$$

The cross-spin terms are new to QGD: body 2's spin drags spacetime via body 1's mass field and vice versa. Explicitly:

$$\sigma_t^{(1)}\sigma_\phi^{(2)} = \alpha_2\sin\theta\,\frac{2G\sqrt{M_1 M_2}}{c^2}\sqrt{\frac{r_1}{\mathcal{S}_1\, r_2\, \mathcal{S}_2}} \tag{13}$$

2.5 Cross-term expansion: $g_{\phi\phi}$ (spin-spin coupling)

$$g_{\phi\phi} = r^2\sin^2\theta\left[1 + (\sigma_\phi^{(1)})^2 + (\sigma_\phi^{(2)})^2 + 2\sigma_\phi^{(1)}\sigma_\phi^{(2)}\right] \tag{14}$$

with spin-spin cross:

$$2\sigma_\phi^{(1)}\sigma_\phi^{(2)} = \frac{4G\alpha_1\alpha_2\sin^2\theta\sqrt{M_1 M_2}}{c^2}\sqrt{\frac{1}{r_1\mathcal{S}_1\, r_2\mathcal{S}_2}} \tag{15}$$

2.6 Equations of motion

In the weak-field limit, the geodesic gives:

$$\boxed{\ddot{\mathbf{x}} = \underbrace{-\frac{GM_1}{r_1^2}\hat{\mathbf{r}}_1 - \frac{GM_2}{r_2^2}\hat{\mathbf{r}}_2}_{\text{Newtonian}} + G\sqrt{M_1 M_2}\left[\frac{\hat{\mathbf{r}}_2}{\sqrt{r_1}\, r_2^{3/2}} + \frac{\hat{\mathbf{r}}_1}{\sqrt{r_2}\, r_1^{3/2}}\right]_{\text{QGD cross}}} \tag{16}$$

The cross-term force scales as $r^{-3/2}$ (softer than Newton) and involves $\sqrt{M_1 M_2}$.

Mutual two-body force. Body 1 moves on the geodesic of body 2's field alone: $\ddot{\mathbf{x}}_1 = -c^2 A_2(\mathbf{x}_1)\nabla A_2(\mathbf{x}_1) = -GM_2/r_{12}^2\,\hat{\mathbf{r}}_{12}$ — exact Newton. Numerically verified: $|a_{\text{QGD}}/a_{\text{Newton}}| = 1.000000$ at all separations $d \geq 2r_s$. The cross-term force appears only when a third body is present.

2.7 Merger condition

The combined event horizon is $g_{tt} = 0$, i.e. $A_{\text{tot}} = 1$. At the saddle point $\mathrm{d}\Sigma/\mathrm{d}r_1 = 0$:

$$\frac{r_1}{r_2} = \left(\frac{M_1}{M_2}\right)^{1/3} \tag{17}$$

Setting $\Sigma = 1$ at the saddle:

$$\boxed{d_{\text{merge}} = \frac{2GM_1}{c^2}\left(1 + \left(\frac{M_2}{M_1}\right)^{1/3}\right)^3} \tag{18}$$

For equal masses $M_1 = M_2 = m$: $d_{\text{merge}} = 16Gm/c^2 = 8r_s^{()} = 4r_s^{(\text{total})}$. Numerically verified: $\Sigma_{\text{saddle}} = 1.000000$ for all mass ratios $q = 0.01$ to $q = 1$.

Table 1: Merger condition at various mass ratios ($M_1 = 10\,M_\odot$)

q	η	$d/r_s^{(\text{tot})}$	Σ_{saddle}	Exact?
1.000	0.2500	4.0000	1.0000000000	Yes
0.500	0.2222	3.8473	1.0000000000	Yes
0.250	0.1600	3.4643	1.0000000000	Yes
0.100	0.0826	2.8535	1.0000000000	Yes
0.010	0.0098	1.7778	1.0000000000	Yes

3 The Three-Body Problem

3.1 Field superposition and metric

For three bodies: $A_{\text{tot}} = \sigma_t^{(1)} + \sigma_t^{(2)} + \sigma_t^{(3)}$, $B_{\text{tot}} = \sigma_\phi^{(1)} + \sigma_\phi^{(2)} + \sigma_\phi^{(3)}$.

3.2 Fully expanded g_{tt}

$$g_{tt} = -\left[1 - \underbrace{\frac{2GM_1 r_1}{c^2\mathcal{S}_1} - \frac{2GM_2 r_2}{c^2\mathcal{S}_2} - \frac{2GM_3 r_3}{c^2\mathcal{S}_3}}_{\text{3 self-energy terms}} - \underbrace{2\sigma_t^{(1)}\sigma_t^{(2)} + 2\sigma_t^{(1)}\sigma_t^{(3)} + 2\sigma_t^{(2)}\sigma_t^{(3)}}_{\text{3 mass cross-terms}}\right] \tag{19}$$

3.3 Fully expanded $g_{t\phi}$

$$\frac{g_{t\phi}}{-r\sin\theta} = \underbrace{\sigma_t^{(1)}\sigma_\phi^{(1)} + \sigma_t^{(2)}\sigma_\phi^{(2)} + \sigma_t^{(3)}\sigma_\phi^{(3)}}_{\text{3 self frame-drag}} + \underbrace{\sigma_t^{(1)}\sigma_\phi^{(2)} + \sigma_t^{(2)}\sigma_\phi^{(1)} + \sigma_t^{(1)}\sigma_\phi^{(3)} + \sigma_t^{(3)}\sigma_\phi^{(1)} + \sigma_t^{(2)}\sigma_\phi^{(3)} + \sigma_t^{(3)}\sigma_\phi^{(2)}}_{\text{6 cross frame-drag}} \tag{20}$$

The three-body metric has $6 + 9 + 6 = 21$ terms total.

3.4 Application: PSR J0337+1715

A millisecond pulsar hierarchical triple (Ransom et al. 2014): $M_1 = 1.4378\,M_\odot$ (pulsar), $M_2 = 0.19751\,M_\odot$ (inner WD), $M_3 = 0.4101\,M_\odot$ (outer WD), $P_{\text{inner}} = 1.629$ d, $P_{\text{outer}} = 327.3$ d.

Predicted orbital decay from energy balance:

$$\frac{\mathrm{d}a_{12}}{\mathrm{d}t} = -\frac{64G^3M_1M_2(M_1+M_2)}{5c^5a_{12}^3}, \qquad t_{\text{merge}} = \frac{5c^5a_{12}^4}{256G^3M_1M_2(M_1+M_2)} \approx 720\ \text{Gyr} \tag{21}$$

The period decay rate $\dot{P}_{\text{inner}}/P_{\text{inner}} \approx -1.7\times10^{-20}$ s/s is measurable with decades of pulsar timing. The inner binary GW frequency $f_{\text{GW}} = 1.4\times10^{-5}$ Hz (LISA band) creates a multi-frequency signal with beat frequencies.

4 The N-Body Metric

For N Kerr bodies $\{M_a, \alpha_a, \mathbf{x}_a(t)\}_{a=1}^N$, define $A_a \equiv \sigma_t^{(a)}$, $B_a \equiv \sigma_\phi^{(a)}$.

$$g_{tt} = -\left[1 - \left(\sum_{a=1}^N A_a\right)^2\right] = -\left[1 - \sum_{a=1}^N A_a^2 - 2\sum_{1\le a<b\le N} A_aA_b\right] \tag{22}$$

$$g_{t\phi} = -r\sin\theta\left(\sum_a A_a\right)\left(\sum_b B_b\right) = -r\sin\theta\left[\sum_a A_aB_a + \sum_{\substack{a,b\\a\neq b}} A_aB_b\right] \tag{23}$$

$$g_{\phi\phi} = r^2\sin^2\theta\left[1 + \sum_a B_a^2 + 2\sum_{a<b} B_aB_b\right], \qquad g_{rr} = -1/g_{tt} \tag{24}$$

Table 2: Cross-term count by number of bodies

N	g_{tt} terms	$g_{t\phi}$ terms	$g_{\phi\phi}$ terms	Total
1	1	1	1	3
2	3	4	3	10
3	6	9	6	21
5	15	25	15	55
10	55	100	55	210
N	$N(N+1)/2$	N^2	$N(N+1)/2$	$N^2+N(N+1)$

Computational complexity: evaluating the metric requires N field evaluations + 2 sums + 2 squarings = $\mathcal{O}(N)$ per field point. Contrast with numerical GR: $\mathcal{O}(N_{\text{grid}}^3 \times N_{\text{steps}})$ with $N_{\text{grid}} \sim 10^9$.

Algorithm:

1. For each body a: evaluate A_a, B_a from (1)–(2).

2. Sum: $A_{\rm tot} = \sum A_a$, $B_{\rm tot} = \sum B_a$.
3. Assemble: $g_{tt} = -(1 - A_{\rm tot}^2)$, etc.
4. To add charge: subtract $\sigma_t^{(Q)}$. To add Λ: add Hr/c.

5 The Superposition Theorem

Theorem 5.1 (N-body superposition)**.** *For N sources with $\varepsilon_{ab} = r_{\rm s}^{(a)} r_{\rm s}^{(b)}/d_{ab}^2 \ll 1$:*

$$\boxed{\frac{|\sigma_t^{(\rm tot)} - \sum_a \sigma_t^{(a)}|}{\sum_a \sigma_t^{(a)}} \le C \max_{a\neq b} \frac{r_{\rm s}^{(a)} r_{\rm s}^{(b)}}{d_{ab}^2}} \tag{25}$$

Proof. **Step 1 (Background expansion).** Write $\sigma = \sigma^{(0)} + \sigma^{(1)} + \cdots$ where $\sigma^{(0)} = \sum_a \sigma^{(a)}$ is the linearised superposition. The operator $\Box_g = \Box + \delta\Box$ where $\delta\Box \sim h^{\mu\nu}\partial_\mu\partial_\nu$ and $h_{\mu\nu}^{(b)} \sim r_{\rm s}^{(b)}/r_b$.

Step 2 (First-order correction). The correction satisfies $\Box\sigma^{(1)} = -\delta\Box\sigma^{(0)} = -\sum_b h_{\mu\nu}^{(b)} \partial^\mu\partial^\nu \sigma^{(0)}$.

Step 3 (Source estimate). Near body a, the dominant contribution from body $b \neq a$:

$$|\delta\Box\sigma^{(a)}| \sim \frac{r_{\rm s}^{(b)}}{d_{ab}^2} \times |\partial\partial\sigma^{(a)}| \sim \frac{r_{\rm s}^{(b)}}{d_{ab}^2} |\Box\sigma^{(a)}| \tag{26}$$

Step 4 (Inversion). From the Green's function: $|\sigma^{(1)}| \lesssim (r_{\rm s}^{(a)} r_{\rm s}^{(b)}/d_{ab}^2)|\sigma_t^{(a)}|$.

Step 5 (Global bound). Summing over $N(N-1)/2$ pairs: global fractional error $\le C \max_{ab} \varepsilon_{ab}$. □ □

Table 3: Superposition accuracy for $10 + 8\,M_\odot$ binary

d/r_s	ε	Accuracy
2	2.0×10^{-1}	80.0%
5	3.2×10^{-2}	96.8%
10	8.0×10^{-3}	99.2%
50	3.2×10^{-4}	99.97%
100	8.0×10^{-5}	99.992%
1000	8.0×10^{-7}	$> 99.9999\%$

Corollary 5.2 (Cross-term exactness)**.** *The metric cross-term $2\sigma_t^{(1)}\sigma_t^{(2)}$ is the exact leading-order contribution from the bilinear structure $g_{tt} = -(1 - \Sigma_{\rm tot}^2)$. It is not an artifact of the approximation; the only approximation is the linearisation of the σ-fields themselves.*

6 The Post-Newtonian Framework

6.1 Notation

For a binary with masses M_1, M_2:

$$M = M_1 + M_2, \quad \mu = M_1 M_2/M, \quad \eta = M_1 M_2/M^2 \;\; (0 < \eta \le 1/4) \tag{27}$$

$$q = M_2/M_1, \quad \delta = (M_1 - M_2)/M = \sqrt{1 - 4\eta} \tag{28}$$

The PN parameter: $x = (GM\Omega/c^3)^{2/3} \approx v^2/c^2$. Circular-orbit relations:

$$r_{12} = \frac{GM}{c^2 x}, \quad \Omega = \frac{c^3}{GM} x^{3/2}, \quad f_{\rm GW} = \frac{c^3}{\pi GM} x^{3/2} \tag{29}$$

Valid for $x \ll 1$; at the Schwarzschild ISCO: $x_{\rm ISCO} = 1/6 \approx 0.167$.

6.2 Conservative sector: two-body equivalence

Theorem 6.1 (QGD conservative sector). *The QGD conservative Hamiltonian for the two-body problem is identical to GR at all PN orders:*

$$H_{\text{QGD}}^{(2-\text{body})} = H_{\text{GR}}^{(2-\text{body})} + \mathcal{O}(G^2\hbar^2/c^4) \tag{30}$$

Proof. The cross-term potential $\delta V_{\text{cross}} = -G\sqrt{M_1 M_2}(\sqrt{r_1}^{-1} r_2^{-3/2} + \sqrt{r_2}^{-1} r_1^{-3/2})$ acts on a test particle at $\mathbf{x}$ in the combined field. In the reduced two-body problem, the cross-term produces exactly the standard GR PN terms at each order; the quantum stiffness contributes $\mathcal{O}(\ell_Q^2/r^2) \sim 10^{-140}$. □

6.3 Binding energy: 1PN to 4PN

$$E(x) = -\frac{\mu c^2}{2} x \left\{ 1 + \sum_{n=1}^{4} e_n x^n + \mathcal{O}(x^5) \right\} \tag{31}$$

Complete coefficients (Blanchet, harmonic gauge):

$$e_1 = -\frac{3}{4} - \frac{\eta}{12} \tag{32}$$

$$e_2 = -\frac{27}{8} + \frac{19}{8}\eta - \frac{\eta^2}{24} \tag{33}$$

$$e_3 = -\frac{675}{64} + \left(\frac{34445}{576} - \frac{205\pi^2}{96} \right) \eta - \frac{155}{96}\eta^2 - \frac{35}{5184}\eta^3 \tag{34}$$

$$\begin{aligned} e_4 = &-\frac{3969}{128} + \left(\frac{123671}{5760} - \frac{9037\pi^2}{1536} + \frac{896\gamma_E}{15} + \frac{448 \ln 16}{15} \right) \eta \\ &- \left(\frac{498449}{3456} - \frac{3157\pi^2}{576} \right) \eta^2 + \frac{301}{1728}\eta^3 + \frac{77}{31104}\eta^4 \end{aligned} \tag{35}$$

Numerical values at the ISCO ($x = 1/6$):

Table 4: Binding energy at ISCO (units of $M_\odot c^2$)

System	η	Newt	1PN	2PN	3PN	4PN
$10 + 10\,M_\odot$	0.250	−0.417	−0.363	−0.331	−0.329	−0.328
$15 + 5\,M_\odot$	0.188	−0.313	−0.273	−0.247	−0.243	−0.239
$36 + 29\,M_\odot$	0.247	−1.338	−1.167	−1.063	−1.057	−1.052

6.4 GW flux: complete coefficients to 3.5PN

$$\mathcal{F}(x) = \frac{32}{5}\frac{c^5}{G}\eta^2 x^5 \left\{ 1 + \sum_n f_n x^n \right\} \tag{36}$$

$$f_1 = -\frac{1247}{336} - \frac{35}{12}\eta \tag{37}$$

$$f_{3/2} = 4\pi \qquad \text{(tail)} \tag{38}$$

$$f_2 = -\frac{44711}{9072} - \frac{9271}{504}\eta - \frac{65}{18}\eta^2 \tag{39}$$

$$f_{5/2} = -\left(\frac{8191}{672} + \frac{535}{24}\eta\right)\pi \tag{40}$$

$$f_3 = \frac{6643739519}{69854400} + \frac{16\pi^2}{3} - \frac{1712\gamma_E}{105} - \frac{856}{105}\ln(16x) + \eta\text{-terms} \tag{41}$$

$$f_{7/2} = \left(-\frac{16285}{504} + \frac{214745}{1728}\eta + \frac{193385}{3024}\eta^2\right)\pi \tag{42}$$

6.5 TaylorF2 phase coefficients

$$\Psi(f) = 2\pi f t_c - \varphi_c - \frac{\pi}{4} + \frac{3}{128\eta x^{5/2}}\left\{1 + \sum_n \psi_n x^n\right\} \tag{43}$$

$$\psi_1 = \frac{3715}{1008} + \frac{55}{12}\eta, \qquad \psi_{3/2} = -10\pi \tag{44}$$

$$\psi_2 = \frac{15293365}{1016064} + \frac{27145}{1008}\eta + \frac{3085}{144}\eta^2 \tag{45}$$

$$\psi_{5/2} = \left(\frac{38645}{1344} - \frac{65}{16}\eta\right)\pi \tag{46}$$

$$\psi_3 = \frac{12348611926451}{18776862720} - \frac{160\pi^2}{3} - \frac{1712\gamma_E}{21} + \eta\text{-terms} \tag{47}$$

$$\psi_{7/2} = \left(\frac{77096675}{2032128} + \frac{378515}{12096}\eta - \frac{74045}{6048}\eta^2\right)\pi \tag{48}$$

7 The QGD Master Formula

7.1 Derivation

The exact Schwarzschild binding energy for circular geodesics:

$$\frac{E_{\text{bind}}}{\mu} = \frac{1-2x}{\sqrt{1-3x}} - 1 = \sum_{n=1}^{\infty} g_n x^n \tag{49}$$

Expanding $(1-3x)^{-1/2} = \sum_{n=0}^{\infty} a_n x^n$ with $a_n = \binom{2n}{n}(3/4)^n$:

$$g_n = a_n - 2a_{n-1} \tag{50}$$

Using the PN form $E = -\mu c^2 x/2(1 + \sum e_n x^n)$, so $g_1 = -1/2$ and $e_n = -2g_{n+1}$:

Theorem 7.1 (QGD master formula)**.**

$$\boxed{e_n^{(\eta=0)} = -\binom{2n}{n}\left(\frac{3}{4}\right)^n \frac{2n-1}{n+1}} \tag{51}$$

Valid for all $n \geq 1$. Generates exact rational numbers.

Proof. Direct: $e_n = -2g_{n+1} = -2(a_{n+1} - 2a_n)$. Using $a_{n+1} = a_n \cdot 3(2n+1)/[2(n+1)]$: $e_n = -2a_n[3(2n+1) - 4(n+1)]/[2(n+1)] = -a_n(2n-1)/(n+1)$. □

Table 5: PN energy coefficients from the master formula (test-body)

n	$e_n^{(\eta=0)}$	Decimal	Status
1	$-3/4$	-0.750	GR known
2	$-27/8$	-3.375	GR known
3	$-675/64$	-10.547	GR known
4	$-3969/128$	-31.008	GR known
5	$-45927/512$	-89.701	QGD prediction
6	$-264627/1024$	-258.425	QGD prediction
7	$-12196899/16384$	-744.440	QGD prediction
8	$-70366725/32768$	-2147.422	QGD prediction

7.2 Results to 8PN

7.3 Convergence to exact Schwarzschild

Table 6: Relative error of PN series vs. exact (test-body)

x	1PN	2PN	3PN	4PN	5PN	6PN
0.01	0.04%	0.001%	$< 10^{-4}$%	$< 10^{-5}$%	$< 10^{-6}$%	$< 10^{-7}$%
0.05	1.0%	0.16%	0.024%	0.003%	0.0005%	0.0001%
0.10	5.6%	1.7%	0.50%	0.14%	0.041%	0.012%
1/6	27.5%	13.8%	6.7%	3.2%	1.6%	0.7%

7.4 The 5PN logarithmic term

At 5PN, hereditary contributions arise from the tail of the σ-propagator:

$$\delta\sigma^{\text{tail}}(x) = -\frac{2GM_{\text{ADM}}}{c^3}\int_{-\infty}^{t} \ln\left(\frac{t-t'}{2r_0/c}\right)\dot{\sigma}^{\text{inst}}(x,t')\,\mathrm{d}t' \tag{52}$$

Iterating to fifth order gives $\delta e_5^{\log} = +\frac{27}{2}\eta\ln(x)$, consistent with the known partial GR result.

8 The Spinning Master Formula

The Kerr equatorial geodesic: $E/\mu = (1-2u+\chi u^{3/2})/\sqrt{1-3u+2\chi u^{3/2}} - 1$. Expanding in (u,χ) yields three closed-form series.

8.1 Spin-orbit coefficients

From $\mathrm{d}E/\mathrm{d}\chi|_{\chi=0}$:

$$\boxed{e_n^{\text{SO}} = -\frac{(2n+1)!!\,3^n}{2^n\,n!}} \tag{53}$$

entering at $\chi\,u^{n+5/2}$ (half-integer PN orders). Verified: $n = 0, 1, 2$ match known GR results at 1.5PN, 2.5PN, 3.5PN.

8.2 Spin-squared coefficients

From $\frac{1}{2}\mathrm{d}^2E/\mathrm{d}\chi^2|_{\chi=0}$:

$$\boxed{e_n^{\mathrm{SS}} = \frac{(2n+3)!!\,3^n}{3\cdot 2^{n+1}\,n!}} \tag{54}$$

entering at $\chi^2 u^{n+3}$. The factor of 1/3 comes from $\Gamma(5/2) = 3\sqrt{\pi}/4$. Verified: $n = 0 \to 1/2$, $n = 1 \to 15/4$, $n = 2 \to 315/16$ — all match Kerr geodesic exactly.

8.3 Verification against exact Kerr

At $u = 0.05$, comparing the 9-term spinning series against the exact Kerr geodesic energy:

Table 7: Spinning series vs. exact Kerr geodesic

χ	Exact	Series ($n = 9$)	Error
0.0	-0.023813	-0.023813	$< 10^{-4}\%$
0.3	-0.024019	-0.024018	0.0003%
0.5	-0.024146	-0.024146	0.0012%
0.7	-0.024267	-0.024266	0.0033%
0.9	-0.024381	-0.024379	0.0070%

9 QGD-Unique Flux Terms

9.1 The QGD dipole (-1PN)

The σ-field sources a radiation dipole weighted by $\sqrt{M_a}$:

$$\boldsymbol{d}_\sigma = \sum_a \sqrt{M_a}\,\mathbf{x}_a, \qquad P_{\mathrm{dip}} = \frac{c^5}{G}\frac{\eta^2 D}{3}x^4 \tag{55}$$

where $D = (\sqrt{M_1} - \sqrt{M_2})^2/M$.

9.2 The QGD dipole tail (0.5PN — no GR analogue)

The tail of the dipole through the σ-propagator:

$$P_{\mathrm{dip}}^{\mathrm{tail}} = \frac{c^5}{G}\eta^2 D\frac{4\pi}{3}x^{5.5} \tag{56}$$

GR's first half-integer flux is at 1.5PN ($x^{6.5}$). The QGD dipole tail at 0.5PN fills the gap.

9.3 Complete QGD flux series

$$\boxed{\mathcal{F}^{\mathrm{QGD}} = \mathcal{F}^{\mathrm{GR}} + \frac{c^5}{G}\eta^2 D\left[\frac{x^4}{3} - \frac{x^5}{2} + \frac{4\pi x^{5.5}}{3} + \frac{2\pi^2 x^7}{3} + \cdots\right] + \frac{c^5}{G}\eta^2 D_3 f_{\mathrm{oct}} x^6} \tag{57}$$

All terms vanish for $M_1 = M_2$ ($D = D_3 = 0$).
Numerical verification for $M_1 = 15\,M_\odot$, $M_2 = 5\,M_\odot$ ($D = 0.134$):
Equal-mass check: $F_{\mathrm{QGD}}^{\mathrm{dip}}|_{M_1=M_2} = 0.000$ W. ✓

Table 8: QGD flux terms beyond GR, ordered by power of x

Power	PN order	Source	GR analogue?	Factor
x^4	-1PN	Dipole	None	$D/3$
x^5	0PN	Dip.$\times$1PN	Different weight	$-D/2$
$x^{5.5}$	0.5PN	Dipole tail	**None**	$4\pi D/3$
x^6	1PN	QGD octupole	GR oct at x^7	$D_3 f_{\rm oct}$
$x^{6.5}$	1.5PN	GR quad tail	Same	4π
x^7	2PN	Dip. tail-of-tail	None	$2\pi^2 D/3$

Table 9: QGD vs GR flux ($15 + 5\,M_\odot$)

x	$\mathcal{F}_{\rm GR}$ (W)	$F^{\rm dip}_{\rm QGD}$ (W)	$\mathcal{F}_{\rm QGD}$ (W)	$\delta F/F$ (%)
0.005	2.52×10^{40}	3.0×10^{-54}	4.73×10^{40}	+88
0.010	7.94×10^{41}	4.8×10^{-53}	1.50×10^{42}	+89
0.050	2.29×10^{45}	3.0×10^{-50}	4.51×10^{45}	+97
0.100	6.95×10^{46}	4.8×10^{-49}	1.40×10^{47}	+102
0.167	9.33×10^{47}	3.8×10^{-48}	1.85×10^{48}	+99

10 TaylorF2 Phase Corrections

Three QGD-specific corrections to the TaylorF2 phase:

$$\delta\Psi^{(-1\mathrm{PN})} = -\frac{5D}{7168\eta^2} x^{-7/2} \propto f^{-7/3} \qquad \text{(unique to QGD)} \tag{58}$$

$$\delta\Psi^{(0.5\mathrm{PN})} = -\frac{5\pi D}{2688\eta^2} x^{-5/2} \propto f^{-5/3} \qquad \text{(same power as GR leading!)} \tag{59}$$

$$\delta\Psi^{(5\mathrm{PN,log})} = -\frac{27\eta}{2} \ln(x) \propto f^0 \ln f \tag{60}$$

The 0.5PN term mimics a chirp-mass shift: $\delta\mathcal{M}_c/\mathcal{M}_c \propto D/\eta^3$.

Table 10: Frequency at which $|\delta\Psi| = 1$ rad ($15 + 5\,M_\odot$)

Term	Scaling	$\lvert\delta\Psi\rvert = 1$ rad	GR analogue
-1PN dipole	$f^{-7/3}$	51 Hz	None
0.5PN dip. tail	$f^{-5/3}$	44 Hz	None
5PN log	$f^0 \ln f$	> 1 kHz	Similar to 3PN

11 Summary

The chapter establishes: (i) exact algebraic N-body metrics with $\mathcal{O}(N)$ complexity; (ii) a superposition theorem with rigorous error bound; (iii) the complete PN hierarchy reproduced identically to GR; (iv) a master formula extending the expansion to arbitrary order with five 5PN–8PN predictions; (v) three spinning series verified against Kerr; and (vi) a complete catalogue of QGD-unique flux terms through 6PN, all vanishing for equal masses.